

Binding Pocket Analysis — Modification Summary

Purpose: replace the relevant C-alpha / Group 3 selections with the custom Binding_Pocket / Group 21 selection, while keeping analyses unchanged when changing the selection would alter their scientific meaning.

Current index definition: Group 21 = Binding_Pocket = 517 atoms.

1. Modified Code Files

1) gmx_2d_interaction_diagram_binding_pocket.py

2) pca_fel_pro_binding_pocket.py

2. Changes in gmx_2d_interaction_diagram_binding_pocket.py

Analysis	Previous selection	New selection	Purpose
Protein RMSD	C-alpha / Group 3	Binding_Pocket / Group 21	Pocket structural stability
Ligand RMSD fit	Fit on C-alpha	Fit on Binding_Pocket	Ligand motion relative to pocket
Protein RMSF	Backbone / Group 4	Binding_Pocket / Group 21	Pocket-residue flexibility
Radius of gyration	Protein / Group 1	Binding_Pocket / Group 21	Pocket compactness/opening
Ligand SASA	Protein + LIG	Binding_Pocket + LIG0	Ligand exposure relative to pocket
PCA	C-alpha / Group 3	Binding_Pocket / Group 21	Collective pocket motions
DCCM / Porcupine	C-alpha	Unchanged	Global protein motion analysis
H-bonds	Protein + LIG	Unchanged	Hydrogen-bond analysis

3. Changes in pca_fel_pro_binding_pocket.py

- Trajectory extraction: Group 21 is extracted instead of C-alpha Group 3.
- Reference: the reference selection is Binding_Pocket.
- Covariance / eigenvectors: calculated from Binding_Pocket motion.
- Cosine content: evaluated for the Binding_Pocket PCA.
- 2D PCA: PC1-PC2, PC1-PC3, and PC2-PC3.
- FEL: generated from the new Binding_Pocket PCA projections.
- 3D PCA: generated from Group 21.
- Extreme structures: extracted using Group 21.
- Cosine vs time: uses Binding_Pocket trajectory segments.
- Holo/Apo: Group 21 must represent the same pocket definition in both systems; a separate Apo index can be supplied when required.

4. Main Commands and Their Purpose

Check the index groups

```
gmx make_ndx -f MD.tpr -n index.ndx
```

Verify that Group 21 is Binding_Pocket.

Exit make_ndx

q

Save/exit when no new group is required.

Calculate RMSF

```
gmx rmsf -s MD.tpr -f MD_center.xtc -n index.ndx -o rmsf.xvg
```

Select Group 21 when prompted.

Run the first analysis script

```
python3 gmx_2d_interaction_diagram_binding_pocket.py --topol MD.tpr --traj MD_center.xtc --index index.ndx --ligand-selection "
```

Run the modified RMSD/RMSF/Rg/SASA/PCA and interaction analyses.

Run PCA + FEL

```
python3 pca_fel_pro_binding_pocket.py --tpr MD.tpr --xtc MD_center.xtc --index index.ndx --group 21 --outdir PCA_out
```

Run PCA, cosine content, projections, FEL, and extreme structures.

5. Expected Results After the Modification

- Ligand RMSD: now describes ligand stability relative to the Binding Pocket rather than relative to the global C-alpha framework.
- Binding-Pocket RMSF: focuses on flexibility of the selected pocket residues.
- Binding-Pocket Rg: describes pocket compactness/opening rather than whole-protein compactness.
- Ligand SASA: describes ligand exposure relative to the Binding Pocket surface.
- PCA/FEL: describe collective motions and conformational states of the Binding Pocket.
- DCCM/Porcupine: remain C-alpha based because they describe broader protein motion; they were not changed mechanically.

6. Important Interpretation Note

The new values are not always numerically comparable 1:1 with the old values. For example, Protein Rg and Binding-Pocket Rg are different physical measurements. For comparisons between simulations, use the same metric and the same selection in every system.

The key scientific change is therefore a shift from a global C-alpha-centered description to a Binding-Pocket-centered description for the selected analyses.

7. Main Output Files

- rmsf.xvg — RMSF of Binding_Pocket when Group 21 is selected.
- RMSD outputs — pocket stability and ligand motion according to the selected fitting group.
- SASA outputs — ligand exposure relative to the pocket.
- PCA_out/scree_plot.html — eigenvalues and cumulative variance.
- PCA_out/cosine_content.html — cosine content.
- PCA_out/proj_12.html, proj_13.html, proj_23.html — PCA projections.
- PCA_out/FEL_12.html, FEL_13.html, FEL_23.html — free-energy landscapes.
- PCA_out/extreme.pdb — extreme structures for visualization.

8. Recommended Workflow

1. Confirm Group 21 in index.ndx: [Binding_Pocket].
2. Run RMSF with -n index.ndx and select 21.
3. Run gmx_2d_interaction_diagram_binding_pocket.py.
4. Run pca_fel_pro_binding_pocket.py with --group 21.
5. Compare Holo/Apo only after confirming that Group 21 represents the same biological pocket in both systems.
6. Interpret the new results as pocket-centered metrics, not as replacements for whole-protein metrics.